

Zomato Restaurant Data Analysis using Python

Internship Task 1: Alfido Tech

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GitHub Link : <https://github.com/ramyarj1307/Zomato-Restaurent-Data-Analysis>

Abstract

This project analyzes the Zomato Restaurant Dataset using Python to understand restaurant ratings, cuisines, and pricing patterns. The dataset was cleaned by removing duplicate records, handling missing values, and converting the required columns into numeric format. Python libraries such as Pandas, Matplotlib, Seaborn, and WordCloud were used to perform Exploratory Data Analysis (EDA). Various visualizations, including a Histogram, Bar Chart, Scatter Plot, Heatmap, and Word Cloud, were created to identify meaningful insights. The analysis showed that most restaurants have good ratings, some cuisines are more popular than others, and restaurant cost has only a moderate relationship with ratings.

Objective

The objective of this project is to analyze the Zomato restaurant dataset using Python and visualize important insights through different charts. This analysis helps understand restaurant ratings, cuisines, costs, and their relationships.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- WordCloud
- Google Colab

Dataset

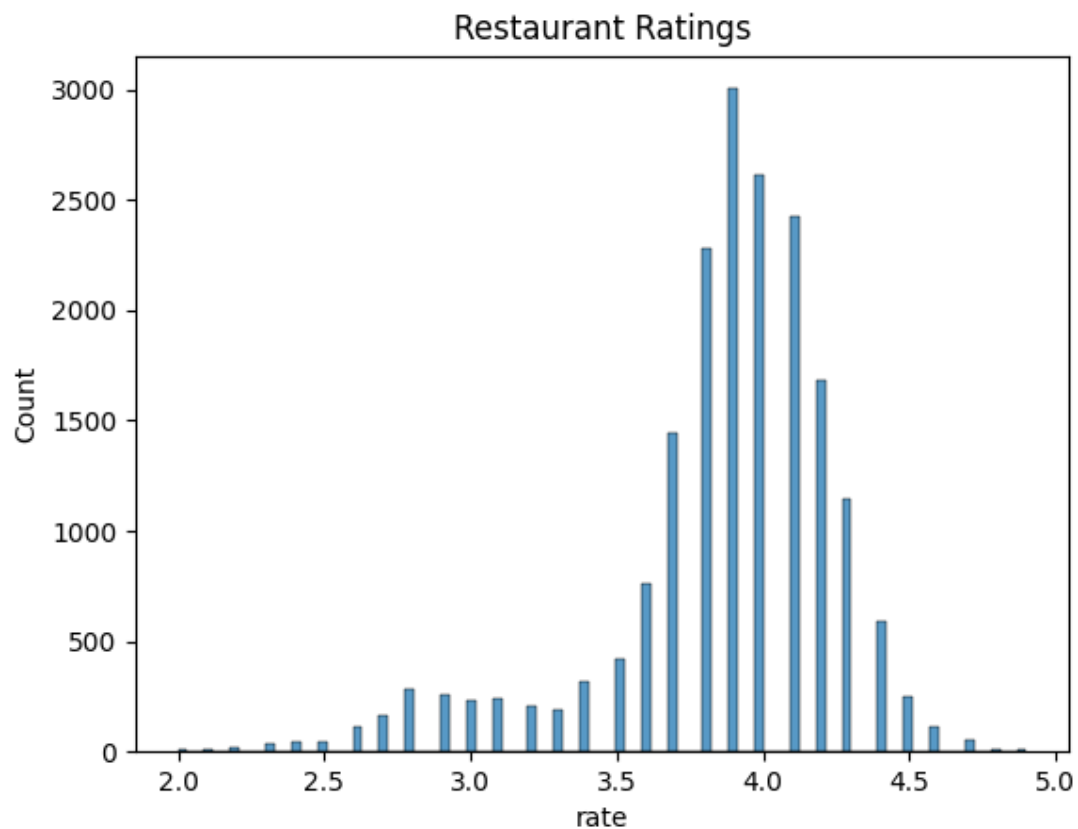
- Dataset Name: Zomato Restaurant Dataset
- Source: Kaggle

Data Cleaning

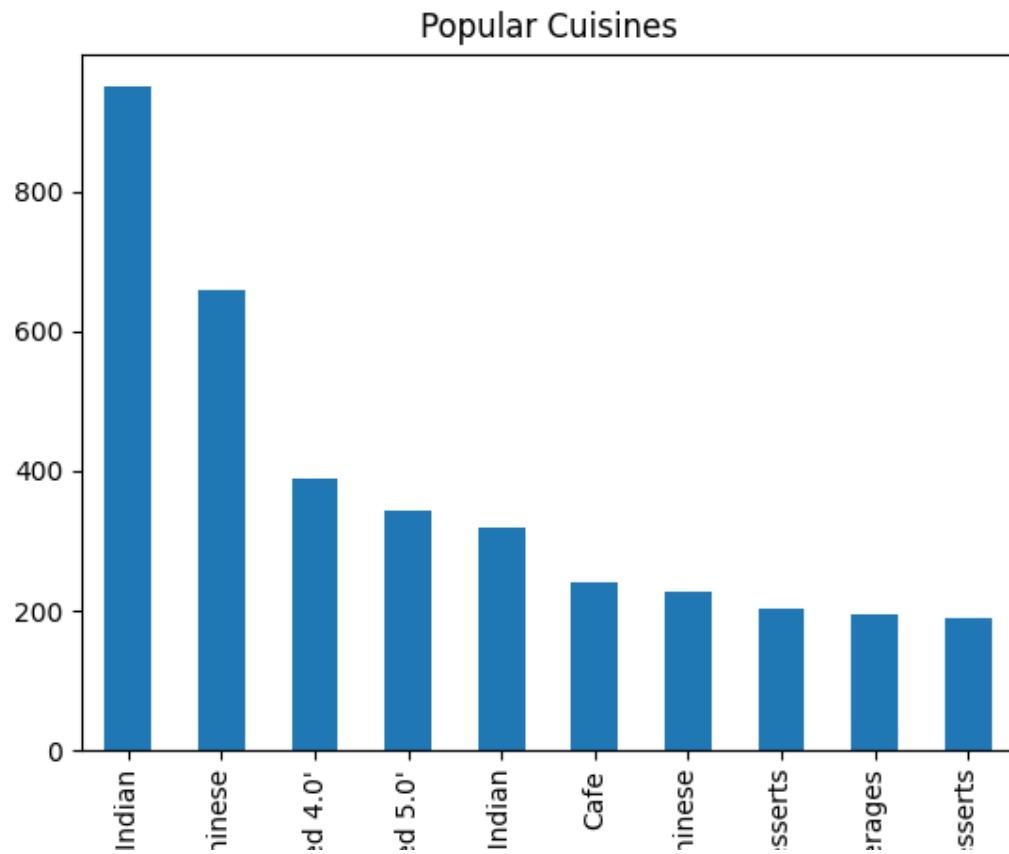
- Loaded the dataset.
- Removed duplicate records.
- Converted the rate column into numeric format.
- Converted the approx cost (for two people) column into numeric format.
- Removed missing values from the required columns.

Visualizations

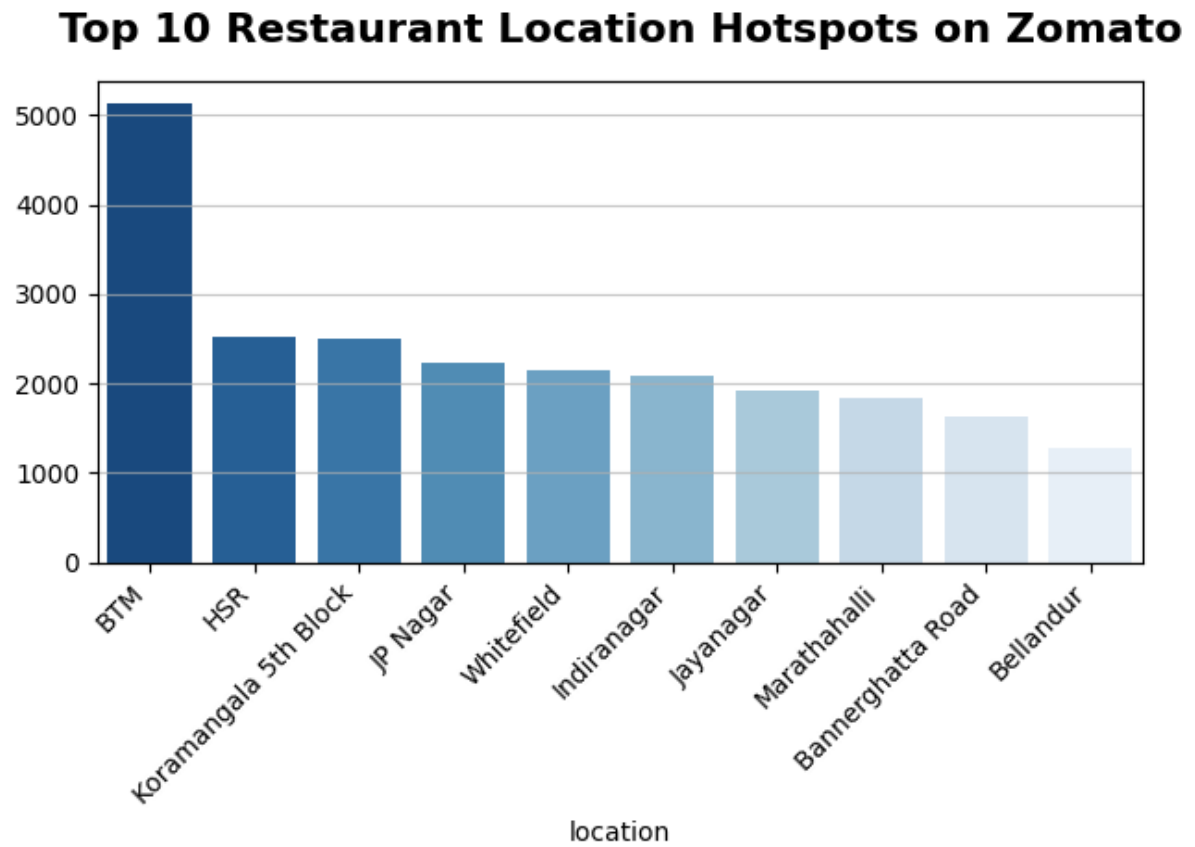
1.Restaurant Ratings Distribution (Histogram)



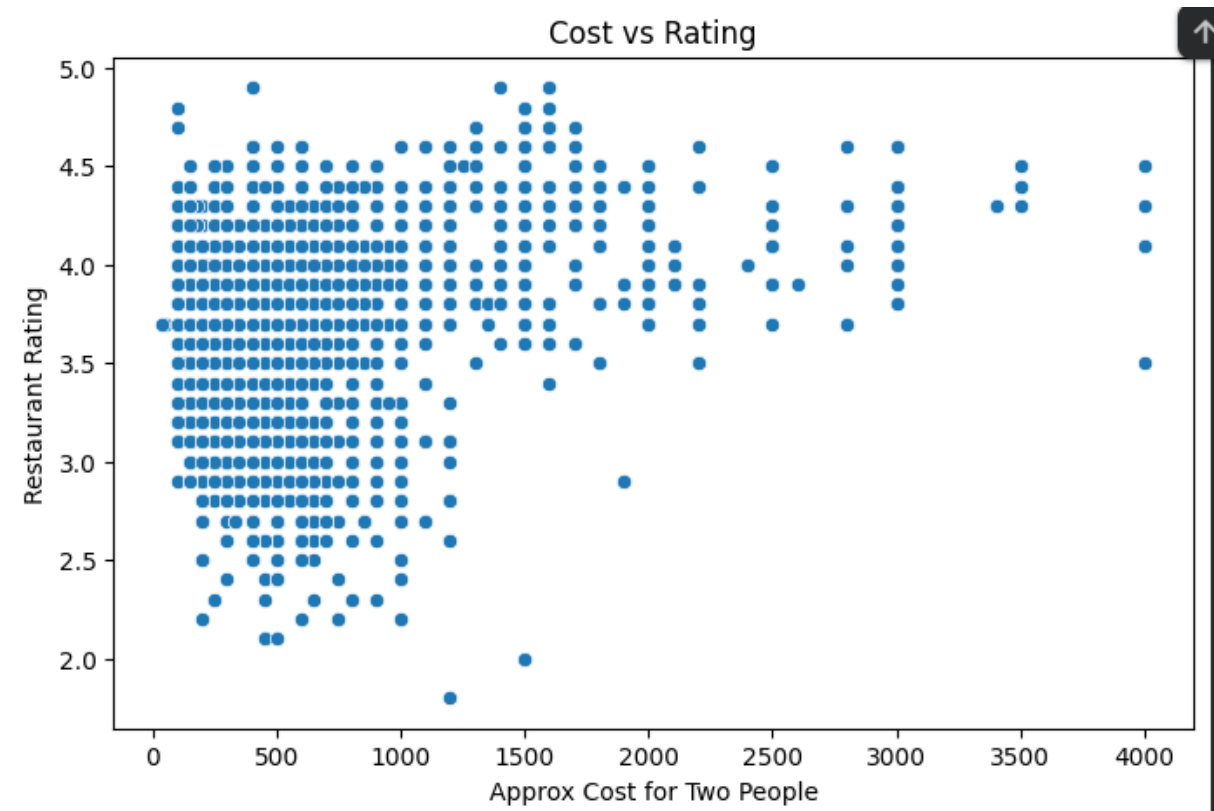
2.Popular Cuisines (Bar Chart)



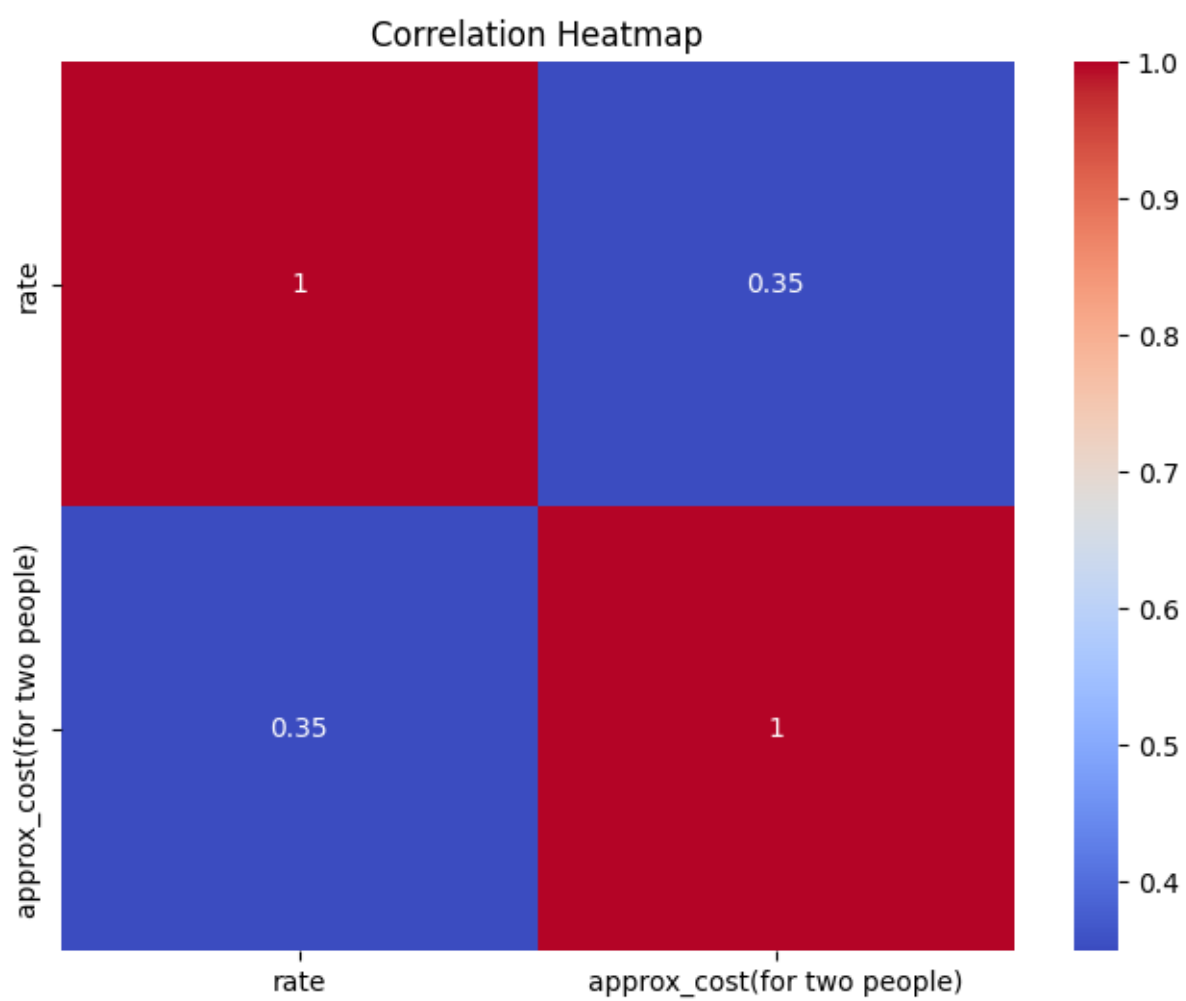
3. Top Restaurant Locations (Bar Chart)



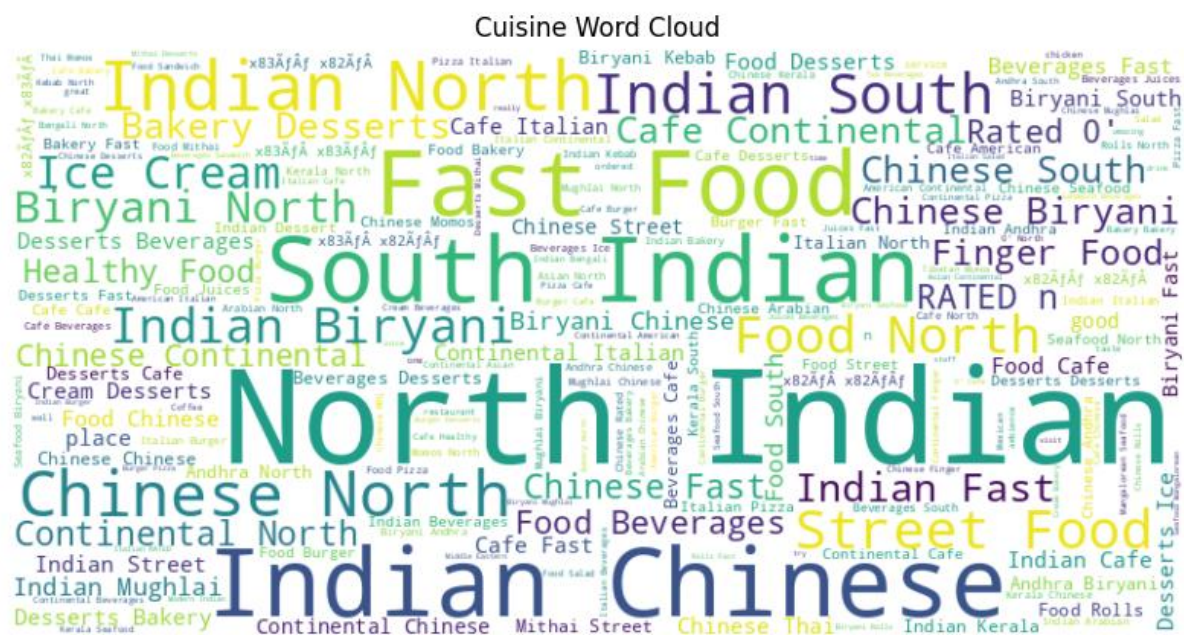
4. Cost vs Rating (Scatter Plot)



5. Correlation Heatmap



6. Word Cloud



Project Summary

Title: Zomato Restaurant Data Analysis using Python

This project analyzes the Zomato restaurant dataset using Python to understand restaurant ratings, cuisines, and pricing. The dataset was imported into Google Colab, and data preprocessing was performed by removing duplicate records, handling missing values, and converting the rate and approximate columns into numeric format.

Exploratory Data Analysis (EDA) was carried out using different visualization techniques. A Histogram was used to study the distribution of restaurant ratings. A Bar Chart showed the most popular cuisines available in the dataset. A Scatter Plot helped analyze the relationship between restaurant cost and ratings. A Heatmap displayed the correlation between numerical features, and a Word Cloud highlighted the most common cuisines or dishes.

The analysis revealed that most restaurants have ratings between 3.5 and 4.5, some cuisines are more popular than others, and restaurant cost has only a moderate relationship with

ratings. Overall, this project demonstrates how Python libraries such as Pandas, Matplotlib, Seaborn, and WordCloud can be used to clean, analyze, and visualize real-world data effectively. The project helped improve data analysis and visualization skills through Exploratory Data Analysis (EDA).

Key Insights

- Most restaurants have ratings between 3.5 and 4.5.
- Some cuisines are more popular than others.
- Restaurant cost has only a moderate relationship with ratings.
- The heatmap shows the correlation between numerical columns.
- The Word Cloud highlights the most common cuisines/dishes in the dataset.

Business Recommendations for Alfido Tech Platform

Based on the exploratory data analysis of the Zomato restaurant dataset, here are 5 strategic business recommendations for an Alfido Tech-style food aggregator or lifestyle platform:

1.Strategic Premium Partnerships with Top-Rated Hotspots

- Insight: The data shows clear regional clusters where certain restaurants maintain excellent ratings.
- Action: The platform should actively approach and onboard these top-rated restaurants as premium partners, offering them featured visibility on the app homepage to drive initial user engagement and trust.

2.Cuisine-Centric Content Marketing and Curated Campaigns

- Insight: Cuisines like Indian and Chinese demonstrate massively higher popularity and customer demand compared to others.

- Action: Launch targeted, cuisine-specific marketing campaigns (e.g., "Biryani Festivals" or "Chinese Fusion Weeks") and create dedicated in-app categories to capture high-volume search traffic.

3.Promote the "Value-for-Money" Segment (High Rating, Low Cost)

- Insight: The scatter plot and correlation analysis ($r = 0.35$) prove that restaurant price does not heavily dictate its quality or rating. High ratings exist even at lower price points.
- Action: Introduce a highly curated section called "Hidden Gems" or "Value-for-Money Hits" to promote highly-rated, budget-friendly restaurants, which will strongly appeal to price-sensitive users like college students.

4.Data-Driven Hyperlocal Expansion into Location Hotspots

- Insight: Specific locations act as dense restaurant hubs with high user activity.
- Action: Focus logistics, delivery infrastructure, and marketing spend aggressively on these identified geographic hotspots to maximize delivery efficiency and reduce customer acquisition costs.

5.B2B Consultation Services for Low-Performing Restaurants

- Insight: A noticeable chunk of restaurants falls into the lower rating bracket (below 3.0) despite being located in high-traffic zones.
- Action: The platform can monetize its data by offering B2B consulting packages to struggling restaurants, guiding them with insights on trending dishes, optimal pricing models, and missing customer expectations to help them improve their ratings.

Conclusion

This project successfully analyzed the Zomato restaurant dataset using Python. Data cleaning and visualization helped identify rating patterns, popular cuisines, and the relationship between restaurant cost and ratings. The project demonstrates how Exploratory Data Analysis (EDA) can extract meaningful insights from real-world datasets.

References: <https://www.kaggle.com/datasets/bhanupratapbiswas/zomato>